Mock exams written by us from the 2026 course materials; they are not official exams. Unofficial solutions, calculations checked by `exams/latex/calc_examens_blancs_embai.py`.

EMBAI, mock midterm 1

50 points, 20 required for the signature. Calculation is required in exercises 2 to 6: a result without calculation earns nothing. Round to 4 decimal places.

Exercise 1 (10 points)

Mark TRUE (T) or FALSE (F). Correct answer +1, wrong answer −1, no answer 0; a negative total is rounded up to 0. No explanation is required or read.

Modelled on: TRUE/FALSE block of the 2026 practice series and past papers

a For large N, an algorithm of complexity $O(N \log_2 N)$ needs less computation than one of complexity $O(N^2)$. T ☐ F ☐

b A binary classifier with 95% accuracy necessarily has a sensitivity above 0.9. T ☐ F ☐

c A decision tree node that contains only positive examples still requires 1 bit of information. T ☐ F ☐

d An attribute with a unique value for every training example would be selected by the information gain criterion, but the resulting tree does not generalize. T ☐ F ☐

e Greedy tree construction based on information gain guarantees the smallest tree consistent with the examples. T ☐ F ☐

f The validation set is used to decide when to stop training; the final performance is measured on a separate test set. T ☐ F ☐

g Bottom-up (agglomerative) hierarchical clustering requires the number of clusters to be specified in advance. T ☐ F ☐

h A multilayer network whose neurons all have linear activation functions is no more powerful than a single-layer network. T ☐ F ☐

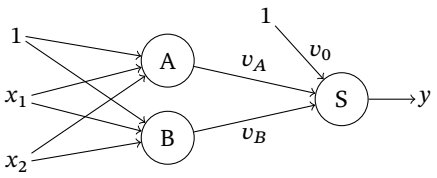
i The derivative of the logistic sigmoid $f(s) = 1/(1+e^{-s})$ never exceeds 0.25. T ☐ F ☐

j An AUC of 0.3 indicates a classification worse than random guessing. T ☐ F ☐

Exercise 2 (9 points)

Modelled on: practice series 1 ex. 2; quiz 06.2 and 06.3; past paper ZH 2023 Q2

The network below has two inputs, two hidden neurons A and B and one output neuron S. All three neurons use the sigmoid $f(s) = 1/(1+e^{-s})$, and each also receives a constant bias input equal to 1. The input $x = (x_1; x_2) = (2; -1)$ is presented with desired output (target) $d = 0.2$. Learning rate $\alpha = 0.5$. Imposed convention: $\varepsilon = d - y$, $E = \frac{1}{2}\varepsilon^2$, $w \leftarrow w - \alpha\, \partial E/\partial w$.



Neuron	bias	x_1	x_2
A	$w_{A0} = 0.2$	$w_{A1} = 0.4$	$w_{A2} = 0.5$
B	$w_{B0} = -0.3$	$w_{B1} = 0.1$	$w_{B2} = 0.3$
	bias	h_A	h_B
S	$v_0 = 0.1$	$v_A = 0.7$	$v_B = -0.5$

a) Forward pass: compute s_A, h_A, s_B, h_B and the output y. (2 pts)

b) Compute $\partial E/\partial v_A$, then the new value of v_A. (2 pts)

c) Compute the new value of w_{A1}, the weight from x_1 to neuron A. (4 pts)

d) Which value of v_A must be used in c), the old or the new one? Is the direction of change of w_{A1} consistent with $y > d$? (1 pt)

Exercise 3 (6 points)

Modelled on: practice series 1 ex. 3

a) Complete the table, then sketch the three functions on $[-3; 3]$, showing asymptotes and slopes. (3 pts)

	$f(-2)$	$f(0)$	$f(2)$	$f'(0)$	$f'(2)$
Sigmoid $1/(1+e^{-s})$					
Tanh $\tanh(s)$					
ReLU $\max(0, s)$					

b) An error must propagate back through four layers of sigmoid neurons to reach the first layer. Keeping only the product of the activation derivatives (weights ignored), give an upper bound of this factor, then compute it if every neuron operates at $s = 2$. Explain the vanishing gradient phenomenon and say which activation functions are most affected. (3 pts)

Exercise 4 (9 points)

Modelled on: practice series 2 ex. 3; past papers ZH 2019 Q2, ZH 2022 Q5

An embedded module predicts whether a pump will fail within a month (Y: yes, N: no) from three attributes: vibration (L low, H high), temperature (C cold, M medium, H hot), age (< 5 years or ≥ 5 years). Twelve examples:

Example	1	2	3	4	5	6	7	8	9	10	11	12
Vibration	L	L	L	H	H	H	L	L	H	H	L	H
Temperature	C	C	C	C	M	M	M	M	H	H	H	H
Age	≥ 5	≥ 5	< 5	< 5	< 5	< 5	≥ 5	< 5	≥ 5	≥ 5	≥ 5	< 5
Failure	N	N	N	N	Y	Y	N	N	Y	Y	Y	N

Hint: $\text{InfoGain}(A) = I\left(\frac{p}{p+n}, \frac{n}{p+n}\right) - \sum_k \frac{p_k+n_k}{p+n} I\left(\frac{p_k}{p_k+n_k}, \frac{n_k}{p_k+n_k}\right)$, with $I(a,b) = -a\log_2 a - b\log_2 b$.

a) How much information is needed at the root? (1 pt)

b) Compute the information gain of each of the three attributes. Which one is tested at the root? (4 pts)

c) For each branch that is not homogeneous, choose the next attribute to test (with the gain). (2 pts)

d) Draw the final tree and check that it classifies the twelve examples correctly. (2 pts)

Exercise 5 (9 points)

Modelled on: practice series 2 ex. 1; past paper ENT 2021 ex. 2; quiz 03.6

An acoustic sensor on a pipeline measures a level p on 5,000 segments, 800 of which have a leak. A segment is classified as "leak" if $p \geq p_{\text{lim}}$. For each threshold, the table gives the number of segments whose level is at or above it. Positive class: leak.

p_{lim}	10	20	30	40	50	60
Segments with a leak, $p \geq p_{\text{lim}}$	800	760	640	400	120	0
Segments without a leak, $p \geq p_{\text{lim}}$	4,200	1,890	630	210	42	0

a) For $p_{\text{lim}} = 30$: give TP, FN, FP, TN, then the sensitivity, specificity, precision, accuracy and F1-score. (4 pts)

b) Compute the (FPR; TPR) point of each threshold and draw the ROC curve. (2 pts)

c) Estimate the AUC with the trapezoidal rule and interpret it. (2 pts)

d) A detector always answers "no leak". Give its accuracy and sensitivity. What does this say about the accuracy found in a)? (1 pt)

Exercise 6 (7 points)

Modelled on: practice series 2 ex. 2; quiz 04.3 and 04.4

Six sensors A to F report the following mean temperatures (°C): A 20.0; B 21.5; C 25.5; D 27.5; E 30.5; F 34.0. The distance between two sensors is the absolute value of the difference. Bottom-up hierarchical clustering is applied.

a) With the minimum (single-linkage) criterion, give the order of the merges and their heights, then draw the dendrogram. (3 pts)

b) With a distance limit of 2.5, how many clusters are obtained? Which ones? (1 pt)

c) Redo the order of the merges with the maximum (complete-linkage) criterion. (2 pts)

d) Do the two criteria give the same partition into two clusters? (1 pt)

EMBAI, mock midterm 2

50 points, 20 required for the signature. Calculation is required in exercises 2 to 6: a result without calculation earns nothing. Round to 4 decimal places.

Exercise 1 (10 points)

Mark TRUE (T) or FALSE (F). Correct answer +1, wrong answer −1, no answer 0; a negative total is rounded up to 0. No explanation is required or read.

Modelled on: TRUE/FALSE block of the 2026 practice series and past papers

a	If one variable is measured in tens of kilograms and another in hundredths of a pH unit, the plain Euclidean distance is dominated by the first one.	T ☐ F ☐
b	Two runs of K-means on the same data can give different clusters if the initial centroids differ.	T ☐ F ☐
c	K-nearest neighbour does most of its computation in the training phase; classifying a new point is then almost free.	T ☐ F ☐
d	PCA can describe a nonlinear relationship between two variables.	T ☐ F ☐
e	The first principal component is the eigenvector of the covariance matrix with the largest eigenvalue.	T ☐ F ☐
f	A sensitivity of 1 guarantees that there is no false positive.	T ☐ F ☐
g	Large neural networks are usually trained directly on the embedded device.	T ☐ F ☐
h	In Edge AI, processing data on the device reduces the response time.	T ☐ F ☐
i	The derivative of ReLU is 0 for $s < 0$ and 1 for $s > 0$.	T ☐ F ☐
j	Changing the weights in the direction of the error gradient, $+\nabla E$, decreases the error.	T ☐ F ☐

Exercise 2 (8 points)

Modelled on: quiz 06.1, 06.2 and 06.3; practice series 1 ex. 2; past papers ZH 2023 Q2, EX 2023 Q4

A neuron computes $s = w_0 + w_1x_1 + w_2x_2 + w_3x_3$ and $y = \tanh(s)$. Current weights: $w_0 = 0.1$; $w_1 = 0.4$; $w_2 = -0.2$; $w_3 = 0.3$. Input $(x_1; x_2; x_3) = (0.5; -1; 2)$, desired output $d = 0.3$, learning rate $\alpha = 0.2$. Imposed convention: $\varepsilon = d - y$, $E = \varepsilon^2$, $w \leftarrow w - \alpha\,\partial E/\partial w$.

a) Compute the output y. (1 pt)

b) Compute $\partial y/\partial w_2$. (2 pts)

c) Compute the new value of w_3. (3 pts)

d) Why not simply use $E = \varepsilon$, or $E = |\varepsilon|$? (2 pts)

Exercise 3 (8 points)

Modelled on: practice series 1 ex. 4; past paper EX 2022 Q2; quiz 04.2

K-means clustering with $K = 2$ and Euclidean distance is applied to the eight points $P_1(3;1)$, $P_2(3;0)$, $P_3(3;4)$, $P_4(4;7)$, $P_5(7;8)$, $P_6(9;8)$, $P_7(9;9)$, $P_8(9;7)$. Initial centroids: $c_1 = (3;5)$, $c_2 = (9;3)$.

a) Assign each point (table of distances) and give the centroids after the first iteration. (3 pts)

b) Perform the second iteration. (2 pts)

c) Perform the third iteration: does the algorithm stop? In which direction did c_2 move at each iteration? (2 pts)

d) With $N = 10{,}000$ points of dimension $d = 8$, $K = 2$ and $J = 20$ iterations, give the order of magnitude of the cost of K-means. (1 pt)

Exercise 4 (8 points)

Modelled on: lectures 3 (K-nearest neighbour) and 4 (distance metrics); quiz 03.1 and 04.1

A motor is described by its current x_1 (A) and its vibration x_2 (mm/s). On historical data the standard deviations are $\sigma_1 = 20$ A and $\sigma_2 = 0.5$ mm/s. Six labelled examples (D: fault, N: normal) and a new motor $q = (60; 2.0)$:

	D1	D2	D3	N1	N2	N3
x_1 (A)	58	63	55	68	48	75
x_2 (mm/s)	3.4	3.1	3.6	2.1	1.8	2.2

a) Compute the Euclidean distance from q to each example and classify q with 3-NN. (3 pts)

b) Repeat with the Mahalanobis distance (diagonal covariance). With 5-NN, what proportion of the neighbours votes for the selected class? Which distance should be used here, and why? (3 pts)

c) The database holds $N = 4{,}000$ examples of dimension $d = 12$. How many multiplications are needed to obtain the N squared Euclidean distances of a new point? Is the square root needed? What is the complexity of one classification? (2 pts)

Exercise 5 (8 points)

Modelled on: lecture 4 (PCA) and lab 3

Four observations of two variables: $(-1;1)$, $(3;1)$, $(5;1)$, $(5;5)$.

a) Centre the data and compute the covariance matrix $\Sigma = \frac{1}{N}\mathbf{X}_{\text{st}}^{\top}\mathbf{X}_{\text{st}}$ ($\mathbf{X}_{\text{st}}$: zero-mean data, one observation per row). (3 pts)

b) Compute the eigenvalues of Σ and the unit eigenvector $\mathbf{u}_1$ of the largest one. (2 pts)

c) What share of the variance does the first component describe? (1 pt)

d) Give the coordinate of each observation on the first component. (1 pt)

e) State one drawback of PCA. (1 pt)

Exercise 6 (8 points)

Modelled on: lecture 5, fall detection use case; lecture 2 (energy)

A wristband detects falls from a three-axis accelerometer sampled at 62.5 Hz, segmented into 2 s windows.

a) A sample is $(a_x; a_y; a_z) = (12; -9; 20)$ m/s². Compute the acceleration magnitude and the G-force. (2 pts)

b) On a window reduced to five values, $X = [10; 9; 10; 15; 11]$, normalize the value 15 as in the lecture (standard deviation computed by dividing by N). (2 pts)

c) How many samples does a window contain, per axis and for the three axes? The 3,600 windows are split 70:15:15; give the size of each set. (1 pt)

d) Anomaly detection, reduced to two dimensions: two clusters with centroids $c_1 = (0; 0)$, $c_2 = (4; 3)$ and boundary radii $r_1 = 1$, $r_2 = 1.5$; anomaly score $d(x) = \min_k(\|x - c_k\| - r_k)$, anomaly if $d(x) > 0.5$. Classify $x_a = (0.6; -0.8)$ and $x_b = (2; 4.5)$. (2 pts)

e) Give two quantified reasons to move the network from float32 to int8, and what must be checked afterwards. (1 pt)

Solutions, mock midterm 1

Exercise 1 (10 points)

Mark TRUE (T) or FALSE (F). Correct answer +1, wrong answer −1, no answer 0; a negative total is rounded up to 0. No explanation is required or read.

Modelled on: TRUE/FALSE block of the 2026 practice series and past papers

a For large N, an algorithm of complexity $O(N \log_2 N)$ needs less computation than one of complexity $O(N^2)$. T ■ F □
$N \log N$ grows more slowly than N^2 (lecture 1, complexity).

b A binary classifier with 95% accuracy necessarily has a sensitivity above 0.9. T □ F ■
Imbalanced classes: a method that always answers "negative" can reach 92% accuracy with zero sensitivity (lecture 3, slide 41).

c A decision tree node that contains only positive examples still requires 1 bit of information. T □ F ■
Homogeneous node: $I(1,0) = 0$ bit.

d An attribute with a unique value for every training example would be selected by the information gain criterion, but the resulting tree does not generalize. T ■ F □
Maximal gain on the training set, no generalization: the tree memorizes the answers (lecture 3, slide 26).

e Greedy tree construction based on information gain guarantees the smallest tree consistent with the examples. T □ F ■
Greedy: a local optimum at each node; finding the smallest tree is in general intractable (lecture 3, slides 13 and 14).

f The validation set is used to decide when to stop training; the final performance is measured on a separate test set. T ■ F □
Lecture 3, slides 31 and 32; lab NN4a, early stopping.

g Bottom-up (agglomerative) hierarchical clustering requires the number of clusters to be specified in advance. T □ F ■
It does not need K; the dendrogram is cut afterwards (lecture 4, slide 28).

h A multilayer network whose neurons all have linear activation functions is no more powerful than a single-layer network. T ■ F □
Without nonlinearity, the capacity reduces to that of a single layer (lab NN4a).

i The derivative of the logistic sigmoid $f(s) = 1/(1+e^{-s})$ never exceeds 0.25. T ■ F □
$f' = y(1-y)$, maximum 0.25 at $s = 0$.

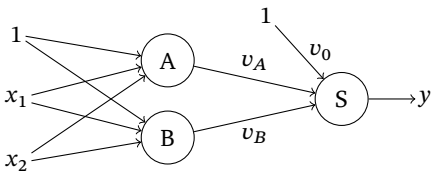
j An AUC of 0.3 indicates a classification worse than random guessing. T ■ F □
AUC < 0.5: worse than random speculation (lecture 3, slide 50).

Key: 6 TRUE, 4 FALSE.

Exercise 2 (9 points)

Modelled on: practice series 1 ex. 2; quiz 06.2 and 06.3; past paper ZH 2023 Q2

The network below has two inputs, two hidden neurons A and B and one output neuron S. All three neurons use the sigmoid $f(s) = 1/(1+e^{-s})$, and each also receives a constant bias input equal to 1. The input $x = (x_1; x_2) = (2; -1)$ is presented with desired output (target) $d = 0.2$. Learning rate $\alpha = 0.5$. Imposed convention: $\varepsilon = d - y$, $E = \frac{1}{2}\varepsilon^2$, $w \leftarrow w - \alpha\, \partial E/\partial w$.



Neuron	bias	x_1	x_2
A	$w_{A0} = 0.2$	$w_{A1} = 0.4$	$w_{A2} = 0.5$
B	$w_{B0} = -0.3$	$w_{B1} = 0.1$	$w_{B2} = 0.3$
	bias	h_A	h_B
S	$v_0 = 0.1$	$v_A = 0.7$	$v_B = -0.5$

a) Forward pass: compute s_A, h_A, s_B, h_B and the output y. (2 pts)

$s_A = 0.2 + 0.4 \cdot 2 + 0.5 \cdot (-1) = 0.5$, $h_A = f(0.5) = 0.6225$.
$s_B = -0.3 + 0.1 \cdot 2 + 0.3 \cdot (-1) = -0.4$, $h_B = 0.4013$.
$s_S = 0.1 + 0.7 \cdot 0.6225 - 0.5 \cdot 0.4013 = 0.3351$, $y = \mathbf{0.5830}$.

b) Compute $\partial E/\partial v_A$, then the new value of v_A. (2 pts)

$\varepsilon = 0.2 - 0.5830 = -0.3830$; $f'(s_S) = y(1-y) = 0.2431$. Local gradient $\delta_S = \varepsilon f'(s_S) = -0.09311$.
$\partial E/\partial v_A = \varepsilon \cdot (-1) \cdot f'(s_S) \cdot h_A = -\delta_S h_A = 0.05796$.
$v_A' = 0.7 - 0.5 \cdot 0.05796 = \mathbf{0.6710}$.

c) Compute the new value of w_{A1}, the weight from x_1 to neuron A. (4 pts)

Hidden local gradient: $\delta_A = f'(s_A)\,\delta_S\,v_A = 0.2350 \cdot (-0.09311) \cdot 0.7 = -0.01532$.
$w'_{A1} = w_{A1} + \alpha\,\delta_A\,x_1 = 0.4 + 0.5 \cdot (-0.01532) \cdot 2 = \mathbf{0.3847}$.
Pitfalls. With $E = \varepsilon^2$ (convention of the past paper solutions) every correction doubles: $v'_A = 0.6420$, $w'_{A1} = 0.3694$. Writing $\varepsilon = y - d$ without changing the update rule reverses the direction. Neuron B plays no role in w_{A1}.

d) Which value of v_A must be used in c), the old or the new one? Is the direction of change of w_{A1} consistent with $y > d$? (1 pt)

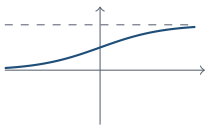
The old one: all gradients are evaluated at the point of the forward pass (with $v'_A = 0.6710$ one would get 0.3853). Direction: $y > d$, so y must decrease; $v_A > 0$ and $x_1 > 0$, hence w_{A1} must decrease: consistent.

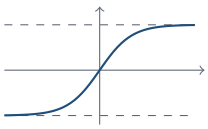
Exercise 3 (6 points)

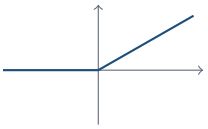
Modelled on: practice series 1 ex. 3

a) Complete the table, then sketch the three functions on $[-3; 3]$, showing asymptotes and slopes. (3 pts)

	$f(-2)$	$f(0)$	$f(2)$	$f'(0)$	$f'(2)$
Sigmoid $1/(1+e^{-s})$	0.119	0.5	0.881	0.25	0.105
Tanh $\tanh(s)$	−0.964	0	0.964	1	0.071
ReLU $\max(0, s)$	0	0	2	undefined	1


Sigmoid, range $(0; 1)$, slope 0.25 at 0


Tanh, range $(-1; 1)$, slope 1 at 0


ReLU, slopes 0 and 1

b) An error must propagate back through four layers of sigmoid neurons to reach the first layer. Keeping only the product of the activation derivatives (weights ignored), give an upper bound of this factor, then compute it if every neuron operates at $s = 2$. Explain the vanishing gradient phenomenon and say which activation functions are most affected. (3 pts)

The gradient of a layer is a product of the derivatives of the layers after it (chain rule). For the sigmoid $f' \le 0.25$, so the factor is at most $0.25^4 = 0.0039$; at $s = 2$, $0.105^4 \approx 1.2 \cdot 10^{-4}$. The first layers receive almost no correction and stop learning. Sigmoid and tanh are the most affected: they saturate ($f' \approx 0$ for large $|s|$). Lab NN4a calls it network paralysis (sigmoidal activation and very large weights). ReLU is little affected: $f' = 1$ for $s > 0$, but $f' = 0$ for $s < 0$.

Exercise 4 (9 points)

Modelled on: practice series 2 ex. 3; past papers ZH 2019 Q2, ZH 2022 Q5

An embedded module predicts whether a pump will fail within a month (Y: yes, N: no) from three attributes: vibration (L low, H high), temperature (C cold, M medium, H hot), age (< 5 years or ≥ 5 years). Twelve examples:

Example	1	2	3	4	5	6	7	8	9	10	11	12
Vibration	L	L	L	H	H	H	L	L	H	H	L	H
Temperature	C	C	C	C	M	M	M	M	H	H	H	H
Age	≥ 5	≥ 5	< 5	< 5	< 5	< 5	≥ 5	< 5	≥ 5	≥ 5	≥ 5	< 5
Failure	N	N	N	N	Y	Y	N	N	Y	Y	Y	N

Hint: $\text{InfoGain}(A) = I\left(\frac{p}{p+n}, \frac{n}{p+n}\right) - \sum_k \frac{p_k+n_k}{p+n} I\left(\frac{p_k}{p_k+n_k}, \frac{n_k}{p_k+n_k}\right)$, with $I(a, b) = -a \log_2 a - b \log_2 b$.

a) How much information is needed at the root? (1 pt)

5 Y, 7 N: $I(\frac{5}{12}, \frac{7}{12}) = \mathbf{0.9799}$ bit.

b) Compute the information gain of each of the three attributes. Which one is tested at the root? (4 pts)

Counts $(p_k; n_k)$ per value; $I(\frac{2}{3}, \frac{1}{3}) = 0.9183$, $I(\frac{1}{6}, \frac{5}{6}) = 0.6500$, $I(\frac{3}{4}, \frac{1}{4}) = 0.8113$.

Attribute	branches	RemainingInfoNeeded	gain
Vibration	H (4; 2), L (1; 5)	0.7842	0.1957
Temperature	C (0; 4), M (2; 2), H (3; 1)	$\frac{1}{3}\cdot 0 + \frac{1}{3}\cdot 1 + \frac{1}{3}\cdot 0.8113 = 0.6038$	**0.3761**
Age	≥ 5 (3; 3), < 5 (2; 4)	0.9591	0.0207

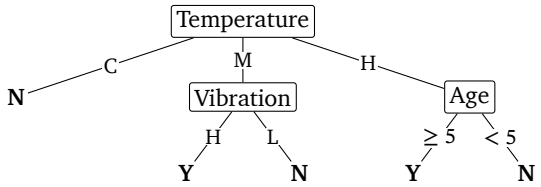
Root: **temperature**. Branch C is a leaf (N).

Pitfalls. Forgetting the weight $\frac{p_k+n_k}{p+n}$ of each branch.

c) For each branch that is not homogeneous, choose the next attribute to test (with the gain). (2 pts)

Branch M (examples 5 to 8, 2 Y, 2 N): vibration splits perfectly, gain 1 (age: 0.3113).
Branch H (examples 9 to 12, 3 Y, 1 N): age splits perfectly, gain 0.8113 (vibration: 0.1226).

d) Draw the final tree and check that it classifies the twelve examples correctly. (2 pts)



All twelve examples reach the correct leaf.

Exercise 5 (9 points)

Modelled on: practice series 2 ex. 1; past paper ENT 2021 ex. 2; quiz 03.6

An acoustic sensor on a pipeline measures a level p on 5,000 segments, 800 of which have a leak. A segment is classified as "leak" if $p \geq p_{\text{lim}}$. For each threshold, the table gives the number of segments whose level is at or above it. Positive class: leak.

p_{lim}	10	20	30	40	50	60
Segments with a leak, $p \geq p_{\text{lim}}$	800	760	640	400	120	0
Segments without a leak, $p \geq p_{\text{lim}}$	4,200	1,890	630	210	42	0

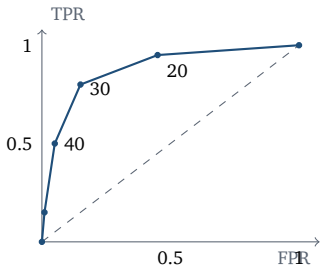
a) For $p_{\text{lim}} = 30$: give TP, FN, FP, TN, then the sensitivity, specificity, precision, accuracy and F1-score. (4 pts)

$P = 800$, $N = 4{,}200$. $TP = 640$, $FN = 160$, $FP = 630$, $TN = 3{,}570$.
Sensitivity (TPR) 640/800 = 0.80; specificity (TNR) 3,570/4,200 = 0.85; precision 640/1,270 = 0.5039; accuracy 4,210/5,000 = 0.842; $F_1 = 2 \cdot 0.5039 \cdot 0.8/1.3039 = 0.6184$.

b) Compute the (FPR; TPR) point of each threshold and draw the ROC curve. (2 pts)

FPR = 1 − specificity = FP/4,200, TPR = TP/800.

p_{lim}	FPR	TPR
10	1	1
20	0.45	0.95
30	0.15	0.80
40	0.05	0.50
50	0.01	0.15
60	0	0



c) Estimate the AUC with the trapezoidal rule and interpret it. (2 pts)

Trapezoids $\frac{(\text{FPR}_{i+1}-\text{FPR}_i)(\text{TPR}_i+\text{TPR}_{i+1})}{2}$: 0.00075 + 0.013 + 0.065 + 0.2625 + 0.53625 = **0.8775**. Well above 0.5 (irrelevant attribute) and below 1.0 (ideal attribute): p is informative.

d) A detector always answers “no leak”. Give its accuracy and sensitivity. What does this say about the accuracy found in a)? (1 pt)

Accuracy 4,200/5,000 = 0.84, sensitivity 0. The detector of a) only reaches 0.842: with imbalanced classes, accuracy alone cannot judge a method; look at sensitivity, precision, F1.

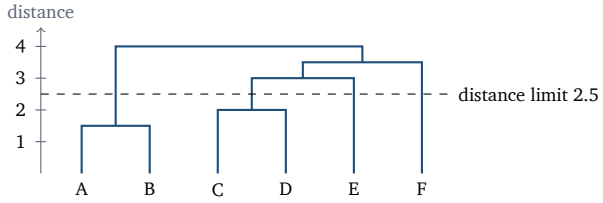
Exercise 6 (7 points)

Modelled on: practice series 2 ex. 2; quiz 04.3 and 04.4

Six sensors A to F report the following mean temperatures (°C): A 20.0; B 21.5; C 25.5; D 27.5; E 30.5; F 34.0. The distance between two sensors is the absolute value of the difference. Bottom-up hierarchical clustering is applied.

a) With the minimum (single-linkage) criterion, give the order of the merges and their heights, then draw the dendrogram. (3 pts)

Consecutive gaps: AB 1.5; BC 4.0; CD 2.0; DE 3.0; EF 3.5. In single linkage, two neighbouring clusters are at the gap between their edges: AB at 1.5; CD at 2.0; (CD)E at 3.0; (CDE)F at 3.5; all at 4.0.



b) With a distance limit of 2.5, how many clusters are obtained? Which ones? (1 pt)

The cut at 2.5 crosses four vertical lines: **4 clusters**, $\{A, B\}$, $\{C, D\}$, $\{E\}$, $\{F\}$.

c) Redo the order of the merges with the maximum (complete-linkage) criterion. (2 pts)

Complete linkage (most distant points): AB at 1.5; CD at 2.0; then (CD)E is 5.0 and EF 3.5: **EF at 3.5**; then (AB)(CD) 7.5 against (CD)(EF) 8.5: **ABCD at 7.5**; all at 14.0.

d) Do the two criteria give the same partition into two clusters? (1 pt)

No: single linkage gives $\{A, B\}$, $\{C, D, E, F\}$ (chaining effect), complete linkage $\{A, B, C, D\}$, $\{E, F\}$. The linkage criterion may lead to different results (quiz 04.4).

Solutions, mock midterm 2

Exercise 1 (10 points)

Mark TRUE (T) or FALSE (F). Correct answer +1, wrong answer −1, no answer 0; a negative total is rounded up to 0. No explanation is required or read.

Modelled on: TRUE/FALSE block of the 2026 practice series and past papers

a If one variable is measured in tens of kilograms and another in hundredths of a pH unit, the plain Euclidean distance is dominated by the first one. T ■ F □
Body mass and blood pH example of lecture 4 (slide 7): use the Mahalanobis distance or scaling.

b Two runs of K-means on the same data can give different clusters if the initial centroids differ. T ■ F □
Sensitive to the initial placement of centroids, local optima (lecture 4, slide 8).

c K-nearest neighbour does most of its computation in the training phase; classifying a new point is then almost free. T □ F ■
No training: the whole cost, $O(dN)$, is paid at each classification (lecture 3, slide 6).

d PCA can describe a nonlinear relationship between two variables. T □ F ■
It can only handle linear relationships (lecture 4, slide 48).

e The first principal component is the eigenvector of the covariance matrix with the largest eigenvalue. T ■ F □
Lecture 4, slides 45 and 46.

f A sensitivity of 1 guarantees that there is no false positive. T □ F ■
Sensitivity = 1 means $FN = 0$; nothing about FP (declaring everything positive gives 1).

g Large neural networks are usually trained directly on the embedded device. T □ F ■
"We usually do not train in embedded devices" (lecture 2, challenge 1); at most a trained model is further adapted.

h In Edge AI, processing data on the device reduces the response time. T ■ F □
Reduced latency through local processing (lecture 1, EdgeAI).

i The derivative of ReLU is 0 for $s < 0$ and 1 for $s > 0$. T ■ F □
Only undefined at 0.

j Changing the weights in the direction of the error gradient, $+\nabla E$, decreases the error. T □ F ■
The step must follow $-\nabla E$.

Key: 5 TRUE, 5 FALSE.

Exercise 2 (8 points)

Modelled on: quiz 06.1, 06.2 and 06.3; practice series 1 ex. 2; past papers ZH 2023 Q2, EX 2023 Q4

A neuron computes $s = w_0 + w_1x_1 + w_2x_2 + w_3x_3$ and $y = \tanh(s)$. Current weights: $w_0 = 0.1$; $w_1 = 0.4$; $w_2 = -0.2$; $w_3 = 0.3$. Input $(x_1; x_2; x_3) = (0.5; -1; 2)$, desired output $d = 0.3$, learning rate $\alpha = 0.2$. Imposed convention: $\varepsilon = d - y$, $E = \varepsilon^2$, $w \leftarrow w - \alpha\,\partial E/\partial w$.

a) Compute the output y. (1 pt)

$s = 0.1 + 0.4 \cdot 0.5 + (-0.2)(-1) + 0.3 \cdot 2 = 1.1$; $y = \tanh(1.1) = \mathbf{0.8005}$.

b) Compute $\partial y/\partial w_2$. (2 pts)

$\partial y/\partial w_2 = f'(s)\,x_2 = (1 - y^2)\,x_2 = 0.3592 \cdot (-1) = \mathbf{-0.3592}$.
Pitfalls. $f' = 1 - y^2$ is computed from y, not from s.

c) Compute the new value of w_3. (3 pts)

$\varepsilon = 0.3 - 0.8005 = -0.5005$. $\dfrac{\partial E}{\partial w_3} = 2\varepsilon \cdot (-1) \cdot f'(s) \cdot x_3 = 2 \cdot 0.5005 \cdot 0.3592 \cdot 2 = 0.7191$.
$w_3' = 0.3 - 0.2 \cdot 0.7191 = \mathbf{0.1562}$. Check: $y > d$ and $x_3 > 0$, so the weight decreases.
Pitfalls. With $E = \frac{1}{2}\varepsilon^2$ the result would be $w_3' = 0.2281$: state the convention.

d) Why not simply use $E = \varepsilon$, or $E = |\varepsilon|$? (2 pts)

$E = \varepsilon$: minimizing it pushes ε towards $-\infty$ (output as large as possible), and over several examples errors of opposite signs cancel out. $E = |\varepsilon|$: not differentiable at $\varepsilon = 0$, and its derivative is ± 1 whatever the size of the error. ε^2 is differentiable everywhere, always positive, and its gradient is proportional to the error.

Exercise 3 (8 points)

Modelled on: practice series 1 ex. 4; past paper EX 2022 Q2; quiz 04.2

K-means clustering with $K = 2$ and Euclidean distance is applied to the eight points $P_1(3;1)$, $P_2(3;0)$, $P_3(3;4)$, $P_4(4;7)$, $P_5(7;8)$, $P_6(9;8)$, $P_7(9;9)$, $P_8(9;7)$. Initial centroids: $c_1 = (3;5)$, $c_2 = (9;3)$.

a) Assign each point (table of distances) and give the centroids after the first iteration. (3 pts)

> Squared distances to $c_1 = (3;5)$ and $c_2 = (9;3)$: P_1 16 and 40; P_2 25 and 45; P_3 1 and 37; P_4 5 and 41; P_5 **25 and 29**; P_6 45 and 25; P_7 52 and 36; P_8 40 and 16.
> Cluster 1: P_1 to P_5; cluster 2: P_6, P_7, P_8. $c_1 = (20/5; 20/5) = (\mathbf{4};\mathbf{4})$, $c_2 = (\mathbf{9};\mathbf{8})$.

b) Perform the second iteration. (2 pts)

> Squared distances to $(4;4)$ and $(9;8)$: P_4 9 and 26 (stays in 1); P_5 25 and 4: P_5 **moves to cluster 2**.
> Cluster 1: P_1 to P_4, $c_1 = (13/4; 12/4) = (\mathbf{3.25};\mathbf{3})$. Cluster 2: P_5 to P_8, $c_2 = (34/4; 32/4) = (\mathbf{8.5};\mathbf{8})$.

c) Perform the third iteration: does the algorithm stop? In which direction did c_2 move at each iteration? (2 pts)

> With $(3.25;3)$ and $(8.5;8)$: P_4 16.56 and 21.25; P_5 39.06 and 2.25; no assignment changes, the centroids no longer move: stop. c_2: from $(9;3)$ to $(9;8)$, upwards by 5; then from $(9;8)$ to $(8.5;8)$, to the left by 0.5.
> *Pitfalls.* Quiz 04.2 method: reassign first, then take the mean; do not guess by eye.

d) With $N = 10{,}000$ points of dimension $d = 8$, $K = 2$ and $J = 20$ iterations, give the order of magnitude of the cost of K-means. (1 pt)

> $O(N \cdot K \cdot J \cdot d)$: $10{,}000 \cdot 2 \cdot 20 \cdot 8 = 3.2 \cdot 10^6$ elementary operations, i.e. $NKJ = 400{,}000$ point to centroid distances.

Exercise 4 (8 points)

Modelled on: lectures 3 (K-nearest neighbour) and 4 (distance metrics); quiz 03.1 and 04.1

A motor is described by its current x_1 (A) and its vibration x_2 (mm/s). On historical data the standard deviations are $\sigma_1 = 20$ A and $\sigma_2 = 0.5$ mm/s. Six labelled examples (D: fault, N: normal) and a new motor $q = (60; 2.0)$:

	D1	D2	D3	N1	N2	N3
x_1 (A)	58	63	55	68	48	75
x_2 (mm/s)	3.4	3.1	3.6	2.1	1.8	2.2

a) Compute the Euclidean distance from q to each example and classify q with 3-NN. (3 pts)

> $d^2 = \Delta x_1^2 + \Delta x_2^2$: D1 5.96; D2 10.21; D3 27.56; N1 64.01; N2 144.04; N3 225.04 (distances 2.44; 3.20; 5.25; 8.00; 12.00; 15.00). Three nearest: D1, D2, D3: **fault**. The current dominates the vibration.

b) Repeat with the Mahalanobis distance (diagonal covariance). With 5-NN, what proportion of the neighbours votes for the selected class? Which distance should be used here, and why? (3 pts)

> $d_M^2 = (\Delta x_1/20)^2 + (\Delta x_2/0.5)^2$: D1 7.85; D2 4.8625; D3 10.3025; N1 0.20; N2 0.52; N3 0.7225. 3-NN: N1, N2, N3: **normal**. 5-NN: N1, N2, N3, D2, D1, i.e. 3 votes out of 5, 60% for "normal". Mahalanobis: the standard deviations differ by a factor 40, so a difference of 1.4 mm/s matters far more than a difference of 2 A (quiz 04.1).

c) The database holds $N = 4{,}000$ examples of dimension $d = 12$. How many multiplications are needed to obtain the N squared Euclidean distances of a new point? Is the square root needed? What is the complexity of one classification? (2 pts)

> $Nd = 48{,}000$ multiplications (as many subtractions, $N(d-1) = 44{,}000$ additions). The square root is useless: it does not change the order of the distances. Cost $O(dN)$ for each classification, with no training phase: heavy for an embedded system, hence parallel computation on several nodes (lecture 3, slide 6).

Exercise 5 (8 points)

Modelled on: lecture 4 (PCA) and lab 3

Four observations of two variables: $(-1;1)$, $(3;1)$, $(5;1)$, $(5;5)$.

a) Centre the data and compute the covariance matrix $\Sigma = \frac{1}{N}\mathbf{X}_{\text{st}}^{\top}\mathbf{X}_{\text{st}}$ ($\mathbf{X}_{\text{st}}$: zero-mean data, one observation per row). (3 pts)

Mean $(3;2)$. Zero-mean data: $(-4;-1)$, $(0;-1)$, $(2;-1)$, $(2;3)$. $\sum x^2 = 24$, $\sum y^2 = 12$, $\sum xy = 8$, hence

$$\Sigma = \frac{1}{4}\begin{pmatrix} 24 & 8 \\ 8 & 12 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}.$$

Pitfalls. Do not forget to centre; divide by $N = 4$ as on the slide.

b) Compute the eigenvalues of Σ and the unit eigenvector $\mathbf{u}_1$ of the largest one. (2 pts)

$\lambda^2 - 9\lambda + 14 = 0$: $\lambda_1 = \mathbf{7}$, $\lambda_2 = \mathbf{2}$. $(6-7)u_x + 2u_y = 0$ gives $u_y = u_x/2$, $\mathbf{u}_1 = (2;1)/\sqrt{5} = (0.8944; 0.4472)$.
Pitfalls. The sign of $\mathbf{u}_1$ is arbitrary.

c) What share of the variance does the first component describe? (1 pt)

Described variance $7/(7+2) = \mathbf{77.8}\%$.

d) Give the coordinate of each observation on the first component. (1 pt)

$\mathbf{u}_1^{\top}\mathbf{x}_{\text{st}}$: $-9/\sqrt{5} = -4.025$; $-1/\sqrt{5} = -0.447$; $3/\sqrt{5} = 1.342$; $7/\sqrt{5} = 3.130$. Check: zero sum, variance $140/20 = 7 = \lambda_1$.

e) State one drawback of PCA. (1 pt)

It can only handle linear relationships; decreased interpretability of the new features (lecture 4, slides 48 and 50).

Exercise 6 (8 points)

Modelled on: lecture 5, fall detection use case; lecture 2 (energy)

A wristband detects falls from a three-axis accelerometer sampled at 62.5 Hz, segmented into 2 s windows.

a) A sample is $(a_x; a_y; a_z) = (12; -9; 20)$ m/s². Compute the acceleration magnitude and the G-force. (2 pts)

$|a| = \sqrt{144 + 81 + 400} = \sqrt{625} = \mathbf{25}$ m/s²; G-force $25/9.8 = \mathbf{2.551}$.

b) On a window reduced to five values, $X = [10; 9; 10; 15; 11]$, normalize the value 15 as in the lecture (standard deviation computed by dividing by N). (2 pts)

Mean 11; deviations $-1, -2, -1, 4, 0$; variance $22/5 = 4.4$; std $= 2.0976$. $x_n = (15-11)/2.0976 = \mathbf{1.907}$.
Pitfalls. Dividing by $N-1$ would give 1.706: follow the formula of the slide.

c) How many samples does a window contain, per axis and for the three axes? The 3,600 windows are split 70:15:15; give the size of each set. (1 pt)

$62.5 \times 2 = 125$ samples per axis, 375 for three axes. Training $0.70 \times 3{,}600 = 2{,}520$; validation 540; test 540.

d) Anomaly detection, reduced to two dimensions: two clusters with centroids $c_1 = (0;0)$, $c_2 = (4;3)$ and boundary radii $r_1 = 1$, $r_2 = 1.5$; anomaly score $d(x) = \min_k(\|x - c_k\| - r_k)$, anomaly if $d(x) > 0.5$. Classify $x_a = (0.6; -0.8)$ and $x_b = (2; 4.5)$. (2 pts)

x_a: $\|x_a - c_1\| = 1$, score $1 - 1 = 0$ (c_2 gives 3.599): $d(x_a) = 0$, **OK**. x_b: $\|x_b - c_1\| - 1 = 3.924$; $\|x_b - c_2\| = \sqrt{4 + 2.25} = 2.5$, score 1.0; $d(x_b) = 1.0 > 0.5$: **anomaly**.
Pitfalls. Take the minimum over the clusters of distance minus radius, not the plain distance.

e) Give two quantified reasons to move the network from float32 to int8, and what must be checked afterwards. (1 pt)

Weight memory divided by 4 (4 bytes against 1). Energy: an 8-bit multiplication costs 0.2 pJ against 3.7 pJ for a 32-bit float multiplication (lecture 2), about 18 times less. Afterwards, compare the accuracy and confusion matrix of the int8 model with those of the float32 model on the same test set: lecture 5 shows a slight drop, with no missed fall.